

HOMEWORK 1
INTRODUCTION TO COMPLEX ANALYSIS
(MATH 4230-40), SPRING 2022

DUE: 01/28/2022

SECTION 1.1 - INTRODUCTION TO COMPLEX NUMBERS

1. Express the following complex numbers in the form $a + bi$:

(a) $(2 + 3i)(4 + i)$

$$\begin{aligned}(2 + 3i)(4 + i) &= 8 + 12i + 2i + 3i^2 \\ &= 8 + 14i - 3 \\ &= 5 + 14i\end{aligned}$$

(b) $(8 + 6i)^2$

$$\begin{aligned}(8 + 6i)^2 &= (8 + 6i)(8 + 6i) \\ &= 64 + 48i + 48i + 36i^2 \\ &= 64 + 96i - 36 \\ &= 28 + 96i\end{aligned}$$

(c) $\left(1 + \frac{3}{1+i}\right)^2$

$$\begin{aligned}\left(1 + \frac{3}{1+i}\right)^2 &= \left(1 + \left(\frac{3}{1+i}\right)\left(\frac{1-i}{1-i}\right)\right)^2 \\ &= \left(1 + \frac{3-3i}{1-i^2}\right)^2 \\ &= \left(1 + \frac{3}{2} - \frac{3}{2}i\right)^2 \\ &= \left(\frac{5}{2} - \frac{3}{2}i\right)^2 \\ &= \frac{25}{4} - \frac{30}{4}i - \frac{9}{4} \\ &= 4 - \frac{15}{2}i\end{aligned}$$

2. Is it true that $\operatorname{Re}(zw) = (\operatorname{Re}z)(\operatorname{Re}w)$? Why or why not?

No, it is not true that $\operatorname{Re}(zw) = (\operatorname{Re}z)(\operatorname{Re}w)$. For $z, w \in \mathbb{C}$, one may note that $z = \operatorname{Re}z + (\operatorname{Im}z)i$ and $w = \operatorname{Re}w + (\operatorname{Im}w)i$, which produces, via direct computation, that

$$\begin{aligned} zw &= (\operatorname{Re}z + (\operatorname{Im}z)i)(\operatorname{Re}w + (\operatorname{Im}w)i) \\ &= \operatorname{Re}z\operatorname{Re}w + (\operatorname{Re}z\operatorname{Im}w)i + (\operatorname{Re}w\operatorname{Im}z)i + (\operatorname{Im}z\operatorname{Im}w)i^2 \\ &= (\operatorname{Re}z\operatorname{Re}w - \operatorname{Im}z\operatorname{Im}w) + (\operatorname{Re}z\operatorname{Im}w + \operatorname{Re}w\operatorname{Im}z)i. \end{aligned}$$

Therefore, we may conclude that

$$\operatorname{Re}(zw) = (\operatorname{Re}z\operatorname{Re}w - \operatorname{Im}z\operatorname{Im}w).$$

Hence, the statement $\operatorname{Re}(zw) = (\operatorname{Re}z)(\operatorname{Re}w)$ is false whenever $\operatorname{Im}z\operatorname{Im}w \neq 0$; i.e., when both $\operatorname{Im}z \neq 0$ and $\operatorname{Im}w \neq 0$.

3. Simplify the following:

(a) $(1+i)^4$

$$\begin{aligned} (1+i)^4 &= (1+2i+i^2)^2 \\ &= (2i)^2 \\ &= -4 \end{aligned}$$

(b) $(-i)^{-1}$

$$\begin{aligned} (-i)^{-1} &= \frac{-1}{i} \\ &= \frac{-1}{i} \left(\frac{-i}{-i} \right) \\ &= \frac{i}{-i^2} \\ &= i \end{aligned}$$

(c) $\frac{1+i}{1-i}$

$$\begin{aligned} \frac{1+i}{1-i} &= \frac{1+i}{1-i} \left(\frac{1+i}{1+i} \right) \\ &= \frac{(1+i)^2}{1-i^2} \\ &= \frac{1+2i+i^2}{2} \\ &= i \end{aligned}$$

4. Simplify the following:

(a) $\sqrt{1 + \sqrt{i}}$

Recall that $\sqrt{a + bi} = \pm(\alpha + \mu\beta i)$, where

$$\alpha = \sqrt{\frac{a + \sqrt{a^2 + b^2}}{2}} \quad \beta = \sqrt{\frac{-a + \sqrt{a^2 + b^2}}{2}}, \quad \mu = \begin{cases} 1 & b \geq 0 \\ -1 & b < 0 \end{cases}$$

We begin by simplifying $\sqrt{i} = \sqrt{0 + (1)i}$ using the above formula:

$$\sqrt{i} = \pm \left(\sqrt{\frac{0 + \sqrt{0^2 + 1^2}}{2}} + (1)\sqrt{\frac{-0 + \sqrt{0^2 + 1^2}}{2}}i \right) = \pm \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right).$$

Therefore, we have

$$\sqrt{1 + \sqrt{i}} = \sqrt{1 \pm \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right)} = \sqrt{\frac{2 \pm \sqrt{2}}{2} + \frac{\sqrt{2}}{2}i}$$

Utilizing the above formula again, we have $\mu = 1$ and

$$\alpha = \sqrt{\frac{\frac{2 \pm \sqrt{2}}{2} + \sqrt{\left(\frac{2 \pm \sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2}}{2}} = \frac{1}{2} \sqrt{2 \pm \sqrt{2} + 2\sqrt{2 \pm \sqrt{2}}}$$

$$\beta = \sqrt{\frac{-\frac{2 \pm \sqrt{2}}{2} + \sqrt{\left(\frac{2 \pm \sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2}}{2}} = \frac{1}{2} \sqrt{-(2 \pm \sqrt{2}) + 2\sqrt{2 \pm \sqrt{2}}}$$

which produces four possible solutions (where $\pm$ and $\mp$ denote the two distinct choices of $+$ or $-$ and remain the same in each instance, e.g. so that $\mp(1 \pm x \pm y)$ denotes $(1 + x + y)$, $(1 - x - y)$, $-(1 + x + y)$, and $-(1 - x - y)$):

$$\sqrt{1 + \sqrt{i}} = \mp \frac{1}{2} \left(\sqrt{2 \pm \sqrt{2} + 2\sqrt{2 \pm \sqrt{2}}} + \left(\sqrt{-(2 \pm \sqrt{2}) + 2\sqrt{2 \pm \sqrt{2}}} \right) i \right).$$

(b) $\sqrt{1 + i}$

$$\sqrt{1 + i} = \pm \left(\sqrt{\frac{1 + \sqrt{1^2 + 1^2}}{2}} + (1)\sqrt{\frac{-1 + \sqrt{1^2 + 1^2}}{2}}i \right) = \pm \left(\sqrt{\frac{1 + \sqrt{2}}{2}} + \sqrt{\frac{\sqrt{2} - 1}{2}}i \right).$$

(c) $\sqrt{\sqrt{-i}}$

We begin by simplifying $\sqrt{-i} = \sqrt{0 + (-1)i}$:

$$\sqrt{-i} = \pm \left(\sqrt{\frac{0 + \sqrt{0^2 + (-1)^2}}{2}} + (-1) \sqrt{\frac{-0 + \sqrt{0^2 + (-1)^2}}{2}} i \right) = \pm \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i \right).$$

During the next step of this computation, care should be taken to note that μ will not remain constant:

$$\sqrt{\sqrt{-i}} = \mp \left(\frac{\sqrt{2 \pm \sqrt{2}}}{2} \pm (-1) \frac{\sqrt{2 \pm (-\sqrt{2})}}{2} i \right)$$

Note that this problem is equivalent to Example 1.1.6 in your textbook.

SECTION 1.2 - PROPERTIES OF COMPLEX NUMBERS

5. Solve the following equations:

(a) $z^5 - 2 = 0$

Solving $z^5 - 2 = 0$ is equivalent to calculating the 5^{th} roots of $2 = 2(\cos 0 + i \sin 0)$, i.e. solving $z^5 = 2$. Using de Moivre's formula, the roots are given by

$$z_k = \sqrt[5]{2} \left(\cos \left(\frac{0}{5} + \frac{2\pi k}{5} \right) + i \sin \left(\frac{0}{5} + \frac{2\pi k}{5} \right) \right), \text{ for } k \in \{0, 1, 2, \dots, 4\}.$$

More exactly, we have:

$$\begin{aligned} z_0 &= \sqrt[5]{2} (\cos(0) + i \sin(0)) = 1 & z_3 &= \sqrt[5]{2} \left(\cos \left(\frac{6\pi}{5} \right) + i \sin \left(\frac{6\pi}{5} \right) \right) \\ z_1 &= \sqrt[5]{2} \left(\cos \left(\frac{2\pi}{5} \right) + i \sin \left(\frac{2\pi}{5} \right) \right) & z_4 &= \sqrt[5]{2} \left(\cos \left(\frac{8\pi}{5} \right) + i \sin \left(\frac{8\pi}{5} \right) \right) \\ z_2 &= \sqrt[5]{2} \left(\cos \left(\frac{4\pi}{5} \right) + i \sin \left(\frac{4\pi}{5} \right) \right) \end{aligned}$$

(b) $z^4 + i = 0$

Solving $z^4 + i = 0$ is equivalent to calculating the 4^{th} roots of $-i = 1(\cos(\pi/2) + i(\pi/2))$, i.e. solving $z^4 = -i$. Using de Moivre's formula, the roots are given by

$$z_k = \sqrt[4]{1} \left(\cos \left(\frac{(\pi/2) + 2\pi k}{4} \right) + i \sin \left(\frac{(\pi/2) + 2\pi k}{4} \right) \right), \quad \text{for } k \in \{0, 1, 2, \dots, 3\}.$$

More exactly, we have:

$$\begin{aligned} z_0 &= \cos \left(\frac{\pi}{8} \right) + i \sin \left(\frac{\pi}{8} \right) & z_2 &= \cos \left(\frac{9\pi}{8} \right) + i \sin \left(\frac{9\pi}{8} \right) \\ z_1 &= \cos \left(\frac{5\pi}{8} \right) + i \sin \left(\frac{5\pi}{8} \right) & z_3 &= \cos \left(\frac{13\pi}{8} \right) + i \sin \left(\frac{13\pi}{8} \right) \end{aligned}$$

6. What is the complex conjugate of $\frac{(3+8i)^4}{(1+i)^{10}}$?

$$\begin{aligned} \overline{\left(\frac{(3+8i)^4}{(1+i)^{10}} \right)} &= \frac{\overline{(3+8i)^4}}{\overline{(1+i)^{10}}} && \text{[By Proposition 1.2.4, part (iii)]} \\ &= \frac{(\overline{3+8i})^4}{(\overline{1+i})^{10}} && \text{[By Proposition 1.2.4, part (ii)]} \\ &= \frac{(3-8i)^4}{(1-i)^{10}}. \end{aligned}$$

This is a sufficiently simplified answer to the problem for our purposes, it's fine if you stopped there. However, note that it is indeed possible to go further using a number of approaches. For example, one of the easiest involves noting that $z^{-1} = \frac{\bar{z}}{|z|^2}$ for $z \neq 0$, which may be employed together with the simple fact that $z^{2n} = (z^2)^n$ to produce:

$$\begin{aligned} \overline{\left(\frac{(3+8i)^4}{(1+i)^{10}} \right)} &= \frac{(3-8i)^4}{(1-i)^{10}} \\ &= (3-8i)^4 \left(\frac{1}{(1-i)} \right)^{10} \\ &= (3-8i)^4 \left(\frac{(1+i)}{|1-i|^2} \right)^{10} \\ &= (3-8i)^4 \left(\frac{(1+i)}{2} \right)^{10} \\ &= \left(\frac{1}{2^5} \right) (3-8i)^4 (1+i)^{10} \\ &= \left(\frac{1}{2^5} \right) ((3-8i)^2)^2 ((1+i)^2)^5 \\ &= \left(\frac{1}{2^5} \right) (-55-48i)^2 (2i)^5 \\ &= \left(\frac{1}{2^5} \right) (721+5280i)(2^5 i) \\ &= -\frac{5280}{32} + \frac{721}{32} i \\ &= -165 + \frac{721}{32} i. \end{aligned}$$

7. Let $w \neq 1$ be an n^{th} root of unity. Show that $1 + w + w^2 + \dots + w^{n-1} = 0$.

Suppose that $w \neq 1$ be an n^{th} root of unity. Write $s = 1 + w + w^2 + \dots + w^{n-1}$. We seek to show that $s = 0$. To wit, note that:

$$\begin{aligned} s &= 1 + w + w^2 + \dots + w^{n-1} \\ sw &= w + w^2 + \dots + w^n \\ sw - s &= (w + w^2 + \dots + w^{n-1} + w^n) - (1 + w + w^2 + \dots + w^{n-1}) \\ s(w - 1) &= w^n - 1 \\ s &= \frac{w^n - 1}{w - 1}. \end{aligned}$$

Note that our assumption $w \neq 1$ is necessary for the final manipulation above. As $w^n = 1$ by our supposition that w is an n^{th} root of unity, it then follows that $s = \frac{1 - 1}{w - 1} = \frac{0}{w - 1} = 0$.

8. Assuming either $|z| = 1$ or $|w| = 1$ and $\bar{z}w \neq 1$, prove that

$$\left| \frac{z - w}{1 - \bar{z}w} \right| = 1.$$

We consider the case where $|z| = 1$ and the case where $|w| = 1$ separately. To begin, suppose first that $|z| = 1$ and $\bar{z}w \neq 1$. Then, we have:

$$\begin{aligned} \left| \frac{z - w}{1 - \bar{z}w} \right| &= \left| \frac{z - w}{\bar{z}z - \bar{z}w} \right| && [\text{Since } \bar{z}z = |z| = 1 \text{ by Proposition 1.2.4, part (iv)}] \\ &= \frac{|z - w|}{|\bar{z}(z - w)|} && [\text{Since } |\xi/\xi'| = |\xi|/|\xi'| \text{ by Proposition 1.2.5, part (ii)}] \\ &= \frac{|z - w|}{|\bar{z}| |z - w|} && [\text{Since } |\xi\xi'| = |\xi| |\xi'| \text{ by Proposition 1.2.5, part (i)}] \\ &= \frac{1}{|\bar{z}|} \\ &= 1 && [\text{Since } |\bar{z}| = |z| = 1 \text{ by Proposition 1.2.5, part (iv)}] \end{aligned}$$

So the result holds when $|z| = 1$ and $\bar{z}w \neq 1$.

Now, we proceed and suppose that $|w| = 1$ and $\bar{z}w \neq 1$. Then, we have:

$$\begin{aligned} \left| \frac{z - w}{1 - \bar{z}w} \right| &= \left| \frac{z - w}{\bar{w}w - \bar{z}w} \right| && [\text{Since } \bar{w}w = |w| = 1 \text{ by Proposition 1.2.4, part (iv)}] \\ &= \frac{|z - w|}{|w(\bar{w} - \bar{z})|} && [\text{Since } |\xi/\xi'| = |\xi|/|\xi'| \text{ by Proposition 1.2.5, part (ii)}] \\ &= \frac{|z - w|}{|w| |(\bar{w} - \bar{z})|} && [\text{Since } |\xi\xi'| = |\xi| |\xi'| \text{ by Proposition 1.2.5, part (i)}] \\ &= \frac{|z - w|}{|(\bar{w} - \bar{z})|} && [\text{Since } |w| = 1 \text{ by assumption}] \\ &= \frac{|z - w|}{|(w - z)|} && [\text{Since } \overline{\xi + \xi'} = \bar{\xi} + \bar{\xi}' \text{ by Proposition 1.2.4, part (i)}] \\ &= \frac{|z - w|}{|(w - z)|} && [\text{Since } |\bar{\xi}| = |\xi| \text{ by Proposition 1.2.5, part (iv)}] \\ &= \frac{|z - w|}{|(-1)(z - w)|} \\ &= \frac{|z - w|}{|(-1)| |z - w|} && [\text{Since } |\xi/\xi'| = |\xi|/|\xi'| \text{ by Proposition 1.2.5, part (ii)}] \\ &= 1 \end{aligned}$$

So the result holds when $|w| = 1$ and $\bar{z}w \neq 1$ as well, thereby completing the proof.